

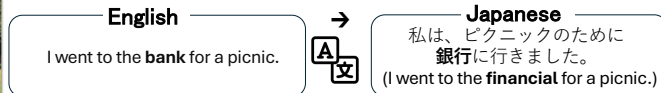
Structured Prompt Design for Translation Agents

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Research background

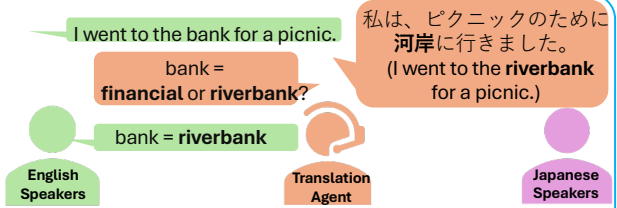
Translation of polysemous or culture-dependent words remains **difficult** in context.

- **Polysemy**: *bank* = financial institution/ riverbank
- **Culture-dependent term**: *public school* = public school (US)/ private school (UK)



Build a **translation agent** that uses **Large Language Models (LLMs)**

- Resolve ambiguity through dialogue with users
- Check context and nuance with users before translating
- Perform translations in a way **similar to human translators**



Research Issue

- Natural language prompts grow longer step by step as dialogue progresses
- This increase in length reduces stability
- It also harms consistency of the model's behavior

Research method

Propose a structured prompt in YAML

- Define **syntax** (form) and **semantics** (meaning) for agent's **scenario**
- Specify three **states**, two **events** (conditions), and three **actions** to control the translation agent's behavior
- Ensure stable behavior across different LLMs
- Provide a model-agnostic, general-purpose framework

Syntax & Semantics

def_scenario:

name: "Specifies the name of the scenario."

states:

- name: "Specifies the name of the state."

rules:

- event: "Specifies the event that triggers the transition."

actions:

- action: "Specifies the name of the action to be executed."

go_state: "Specifies the next state the agent transitions to after the event occurs."

def_action:

name: "Specifies the name of the action."

function: "Describes the functionality of the action."

def_event:

name: "Specifies the name of the event."

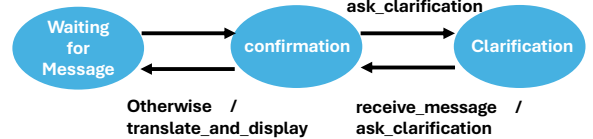
content: "Describes the conditions or triggers for the event."

Event (condition) /
Action (behavior)

*ambiguity : Polysemy, Culture-dependent term

received_message /
check_*ambiguity

*ambiguity_detected
/
ask_clarification



scenario

def_scenario:

name: "Message Clarification and Translation"

states:

- name: "**Waiting for Message**"

- name: "**Confirmation**"

rules:

- event: "receive_message"

- event: "ambiguity_detected"

actions:

- action: "check_ambiguity"

actions:

- action: "ask_clarification"

go_state: "Confirmation"

go_state: "Clarification"

- event: "otherwise"

actions:

- action: "translate_and_display"

go_state: "Waiting for Message"

- name: "**Clarification**"

rules:

- event: "receive_message"

actions:

- action: "check_ambiguity"

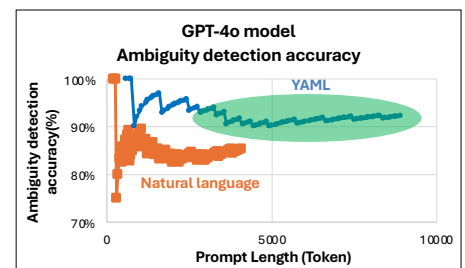
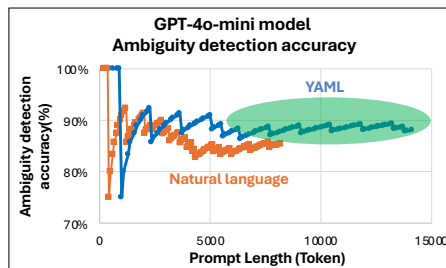
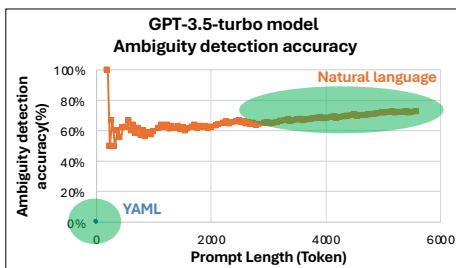
go_state: "Confirmation"

Evaluation results

Experiment Procedure : Using a single initial instruction prompt in each format (YAML-based and **natural language**-based), we translate **100** English sentences successively.

Dataset : **100 English sentences** (50 ambiguous sentences + 50 unambiguous sentences)

Evaluation items : **Ambiguity detection accuracy**: the proportion of cases where the system correctly identified ambiguity and asked the user for clarification.



Conclusion

Structured **YAML** prompts **improve** translation agent consistency on high-performance LLMs.

- **GPT-4o** keeps outputs consistent even with longer prompts
- **GPT-3.5-turbo** works better with natural language prompts due to lower processing load

We will use this translation agent for **dialogue support(honorifics)**.

YAML prompts are robust to token length growth, but we must find ways to **reduce** tokens to avoid high cost.

